Evidence Availability and Retrieval in Scientific Claim Verification: A Matched SciFact Development-Set Study

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ABSTRACT

Scientific claim verification requires both the retrieval of relevant evidence and its correct interpretation. This study compares three evidence-input conditions on a matched subset of the SciFact development split: claim only (A), claim plus annotated gold evidence (B), and claim plus the top five abstracts retrieved using BM25 (C). Each condition uses a separately fitted TF-IDF and logistic-regression classifier trained on the same training split with condition-specific inputs. The archived evaluation contains 286 matched development claims, including 182 claims with annotated gold evidence. On all 286 claims, macro-F1 is 0.3757 for A, 0.4988 for B, and 0.3650 for C. The apparent advantage of B on the full set is confounded because NOT ENOUGH INFO claims have empty evidence, creating an evidence-presence shortcut. On the 182-claim gold-evidence slice, B reaches macro-F1 0.3492 versus 0.3140 for A; the paired difference is +0.0352 (95% bootstrap CI: -0.0275 to 0.0974), which does not satisfy the predeclared improvement rule. BM25 achieves a claim-level Hit@5 of 90.11%, compared with 82.97% for TF-IDF cosine retrieval, but the BM25-based classifier does not outperform the claim-only model on the evidence slice. A 50-case error sample was prepared for taxonomy-guided review, but its AI-assisted codes remain provisional and are not reported as final findings. The results show that retrieval success and correct scientific-claim classification are distinct outcomes, and that the tested lexical verifier remains weak, especially for CONTRADICT claims.

Keywords: SciFact; scientific claim verification; evidence retrieval; BM25; TF-IDF; logistic regression; claim-level Hit@5; macro-F1

1. Introduction

Scientific claims often require evidence from specialized research literature. A verifier must identify relevant research abstracts and determine whether their findings support a claim, contradict it, or leave it unresolved. SciFact formalizes this task using expert-written scientific claims, evidence-containing abstracts, and rationale annotations [1]. Its structure makes it possible to study two related but distinct operations: retrieving evidence and assigning a verdict after evidence has been identified.

These operations should not be treated as interchangeable. A retrieval system may return a relevant paper without the downstream classifier correctly interpreting the relationship between the paper and the claim. Conversely, a classifier may achieve apparently strong performance by exploiting input patterns that correlate with labels rather than by reasoning over the scientific content. Therefore, evidence availability, retrieval quality, and classification performance need to be evaluated separately.

This study examines three evidence-input conditions under a matched SciFact development-set protocol: (A) claim-only input, (B) claim combined with annotated gold evidence, and (C) claim combined with the top five abstracts retrieved by BM25. The implementation uses TF-IDF features and logistic regression, chosen as a lightweight and CPU-friendly baseline. The aim is an auditable comparison of condition-specific pipelines, not a new verification architecture or a claim that oracle-versus-retrieved evidence comparisons are unprecedented. Prior work, including CliVER, already investigates retrieval-augmented scientific claim verification [2].

The paper makes four bounded contributions. First, it reports a per-claim matched comparison of the three input conditions. Second, it reports retrieval success independently from verdict classification, using a claim-level Hit@5 measure. Third, it audits a structural shortcut in the gold-evidence condition caused by empty evidence for NOT ENOUGH INFO claims. Fourth, it documents the limitations of the archived run and the provisional status of the planned taxonomy-guided error review. These contributions are intended to support careful interpretation and reproducibility rather than to establish a new state of the art.

2. Related Work

Wadden et al. introduced SciFact as a benchmark for scientific claim verification that requires systems to retrieve evidence and identify rationales supporting or refuting claims [1]. The benchmark motivates the separation of document retrieval from verdict prediction. Liu et al. proposed CliVER, an end-to-end retrieval-augmented system for scientific claim verification, and evaluated label prediction on SciFact and CoVERt; their work also considers Open and Oracle settings for a separate clinician-claim task [2]. Accordingly, the present study does not claim novelty for comparing gold and retrieved evidence. Its focus is the matched evaluation record, the diagnosis of an empty-evidence shortcut, and a cautious assessment of a lightweight lexical verifier.

Several adjacent evaluation frameworks clarify why evidence retrieval and downstream correctness should be reported separately. FACTSCORE decomposes long-form generated text into atomic facts and estimates the proportion supported by a knowledge source [3]. ALCE evaluates citation-enabled generation using dimensions that include correctness and citation quality [4]. RAGAs separates retrieval-context quality from generation-related evaluation in retrieval-augmented generation [5]. These frameworks motivate fine-grained support analysis, but their tasks and metrics are not numerically interchangeable with three-way SciFact classification.

RAV emphasizes the importance of retrieval quality for fact verification on FEVER [6]. Chen et al. investigate realistic evidence retrieval for complex political claims using temporally constrained web evidence [7]. Two 2026 preprints examine retrieval utility and taxonomy-based error diagnosis in biomedical or medical factuality settings [10, 11]; they are treated here as emerging related work, not as established baselines. BM25 is used as a lexical retrieval method following the probabilistic relevance framework [8]. TF-IDF and logistic regression are implemented using the scikit-learn ecosystem [9].

3. Research Questions and Decision Rule

The study addresses the following research questions:

- RQ1: On the same eligible SciFact development claims, how do claim-only (A), claim plus annotated gold evidence (B), and claim plus retrieved evidence (C) affect three-way verification performance?
- RQ2: How often does top-five retrieval include at least one annotated gold evidence document, and how does retrieval hit/miss status relate descriptively to downstream correctness?
- RQ3: Which prespecified error categories appear in each condition? A 50-case error sample and a frozen codebook were prepared, but the available AI-assisted codes remain provisional pending student verification against the exact historical corpus snapshot.

Macro-F1 is the primary RQ1 metric because it gives equal weight to the classes despite their unequal frequencies. The predeclared practical-effect reference is an absolute macro-F1 gain of +0.03. An improvement is considered supported only when the estimated gain is at least +0.03 and the 95% paired-bootstrap confidence interval excludes zero. The expectations that B would exceed A and that C would approach B are treated as hypotheses to examine, not as findings.

4. Dataset and Data Construction

4.1. Dataset partitions

The archived data audit records 809 labelled training claims, 300 development claims, and 5,183 corpus documents. The official test split contains 300 claims without publicly available labels in the benchmark release and was not used for metric evaluation. The official repository describes the train, development, and test files and directs test evaluation to the benchmark leaderboard [1]. Accordingly, all results in this paper are development-set results; no test-set or leaderboard performance is claimed.

4.2. Matched evaluation set and duplicate filtering

Fourteen development claims with near-duplicate training counterparts were held out using a token-set Jaccard similarity threshold of at least 0.90. This left 286 claims shared across the three evaluation conditions. Their class distribution is 60 CONTRADICT, 104 NOT ENOUGH INFO, and 122 SUPPORT. Of these, 182 claims have annotated gold evidence: 60 CONTRADICT and 122 SUPPORT. The evidence-bearing subset therefore contains two classes and no NOT ENOUGH

INFO examples. A sensitivity analysis using a stricter near-duplicate similarity threshold of 0.70 leaves 209 matched claims, of which 127 have annotated gold evidence.

4.3. Label mapping and evidence construction

The archived label mapping interprets an empty evidence annotation as NOT ENOUGH INFO, one distinct SUPPORT or CONTRADICT annotation as that corresponding label, and a missing evidence key as an unlabelled record. Records with mixed or unknown mappings are excluded. Claim and document identifiers are reconciled before the three conditions are matched. Retrieval scoring excludes claims without annotated gold evidence from its denominator; such claims are not counted as retrieval failures. The mapping was recorded in the project audit outputs and should be rechecked against the exact raw dataset schema on any fresh acquisition.

For condition B, the input uses the first annotated rationale set per evidence document in file order, following a logged protocol correction. An independent audit identified that 75 of the 182 evidence-eligible matched claims have multiple rationale sets in at least one document. Selecting the first rationale set is therefore a consequential operational choice. The results should not be generalized to union-of-rationales or sentence-level evidence aggregation without a new ablation.

5. Experimental Protocol

5.1. Evidence-input conditions

Condition A receives only the scientific claim. Condition B receives the claim together with the first annotated rationale set for each gold evidence document, in file order. NOT ENOUGH INFO claims have empty gold evidence in B; therefore, B is an oracle-style evidence condition only on the 182-claim subset with annotated gold evidence. Condition C receives the claim together with title-plus-abstract text from the top five documents returned by BM25. Gold document identifiers and labels are used only after retrieval for evaluation; they are not used to rank candidate documents.

5.2. Classification model and training controls

Each condition is trained as a separate TF-IDF plus logistic-regression model. All models use the same training claim identifiers, the same algorithm family, the same random seed, and the same core hyperparameters, but their training inputs reflect the corresponding condition. The vectorizer uses word unigrams and bigrams, min_df = 1, L2 normalization, and a maximum of 100,000 features. Logistic regression uses C = 1, the lbfgs solver, a maximum of 2,000 iterations, no class weighting, and argmax prediction. The vocabulary is fitted on training material only. The class order is CONTRADICT, NOT ENOUGH INFO, and SUPPORT.

This design compares end-to-end condition-specific pipelines. It does not hold one fitted classifier fixed and change only the evidence supplied at evaluation time. Consequently, observed differences may reflect changes in the training representation as well as changes in the evidence available at inference. The study does not interpret these differences as a causal effect of evidence availability alone.

5.3. Retrieval methods

BM25 is implemented with rank-bm25 version 0.2.2 using k1 = 1.5, b = 0.75, and epsilon = 0.25. Queries are lowercased and tokenized using the recorded word-character pattern (\w+). No stop-word removal or stemming is applied. The top five documents are concatenated in ranked order using title and abstract text. Ties in retrieval scores are resolved by ascending numeric document identifier. A TF-IDF cosine retriever is used as a second retrieval comparator. Both retrievers operate over the same corpus and query set.

5.4. Evaluation metrics and uncertainty

Classification is evaluated using accuracy, macro-F1, per-class precision, recall and F1, and confusion matrices. Paired differences B-A, C-A, and B-C are estimated with 2,000 claim-level bootstrap resamples using seed 42 and percentile 95% confidence intervals. Conditions remain paired by claim identifier within each resample.

Retrieval is evaluated using claim-level Hit@5 over the 182 claims with annotated gold evidence. A claim is a hit when at least one of its annotated gold evidence documents appears among the five retrieved documents. Thus, Hit@5 is the proportion of eligible claims with one or more gold-document hits. It is not document-level recall, which would require counting the number of relevant documents retrieved relative to the total number of relevant documents.

6. Audit and Protocol Validation

6.1. Rationale for the development-set evaluation

The labelled test split is not publicly distributed in the benchmark release, so the course project uses the official development split. The analysis is explicitly described as a development-set evaluation. The primary matched set uses the 0.90 near-duplicate threshold and retains 286 claims; the 0.70 sensitivity analysis changes both the sample size and included examples. Results from the two thresholds are therefore not direct replications on identical claims.

The full 286-claim set contains all three labels, whereas the 182-claim gold-evidence slice contains only SUPPORT and CONTRADICT. Its macro-F1 is consequently a two-class average, not the same estimand as the three-class macro-F1 on all matched claims. The paper keeps these slices separate when interpreting results.

6.2. Internal consistency checks

The timestamped independent audit reports 12 PASS checks, four WARN checks, two INFO checks, and zero FAIL checks. Checks include agreement of training and development label counts with the SciFact paper, corpus size, the absence of invalid evidence-document references, recomputation of metrics from saved predictions, independent refitting of condition A, train-only vocabulary fitting, independent BM25 spot-checking, retrieval Hit@5 recomputation, and hash consistency between configurations and outputs. These checks support the internal consistency of the archived files; they do not establish that every modeling choice reproduces the official SciFact system.

An independent refit of condition A reported a maximum absolute probability difference of approximately 5 x 10^-9 compared with the saved probabilities, consistent with the effect expected from saving probabilities to eight decimal places. The independent vocabulary check found 9,018 terms when fitting on training data only, compared with 10,957 terms if training and development material were combined. This check would detect development-data contribution to the vocabulary.

7. Results

7.1. RQ1: classification performance

Table 1 summarizes the saved evaluation outputs. On the 286-claim matched set, condition B has the highest macro-F1 (0.4988), followed by A (0.3757) and C (0.3650). On the 182-claim gold-evidence slice, B has macro-F1 0.3492, A has 0.3140, and C has 0.3119. Accuracy follows the same broad ordering for the B condition, but the all-claim result is subject to the empty-evidence shortcut discussed below.

Evaluation slice	Condition	Accuracy	Macro-F1
All matched (n = 286)	A: Claim only	0.4615	0.3757
	B: Gold evidence	0.6014	0.4988
	C: BM25 top-5	0.4371	0.3650
Gold-evidence (n = 182)	A: Claim only	0.3901	0.3140
	B: Gold evidence	0.4835	0.3492
	C: BM25 top-5	0.3736	0.3119

Table 1. Saved development-set classification results. The 286-claim slice is three-class; the 182-claim evidence slice is two-class.

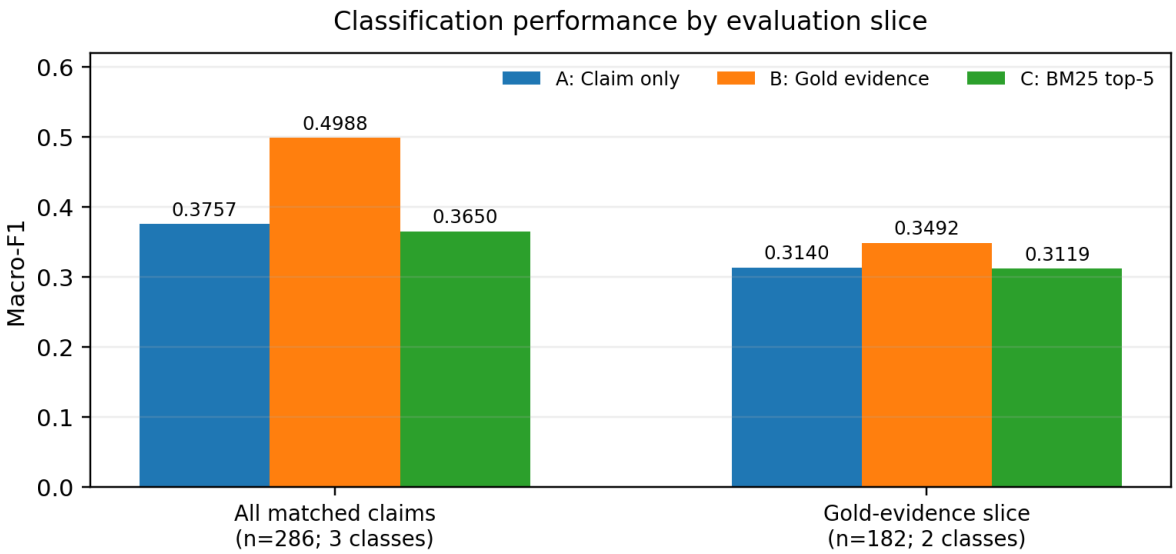


Figure 1. Macro-F1 by evidence-input condition. The two evaluation slices contain different class sets and should not be compared as identical estimands.

On the primary 286 claims, the B-A macro-F1 difference is +0.1231 (95% CI: 0.0721 to 0.1783). This numerically positive difference is not interpretable as a gold-evidence benefit because the empty evidence field encodes the NOT ENOUGH INFO class in the B condition. The evidence-bearing slice is the cleaner available comparison for claims with gold evidence, but it contains only two classes.

On that 182-claim slice, the B-A difference is +0.0352 (95% CI: -0.0275 to 0.0974). Although the point estimate exceeds the +0.03 practical-effect reference, the interval includes zero, so the predeclared rule is not satisfied. The C-A difference is -0.0021 (95% CI: -0.0629 to 0.0591), which is also inconclusive.

7.2. Per-class performance

Table 2 reports per-class F1 on all 286 matched claims. The most pronounced weakness is the CONTRADICT class: F1 is 0.0476 for A, 0.0444 for B, and 0.0659 for C. By comparison, SUPPORT F1 is 0.4929, 0.6078, and 0.4676, respectively. Condition B also has a high NOT ENOUGH INFO F1 of 0.8442, consistent with its ability to exploit the empty-evidence pattern. This score should not be interpreted as evidence-grounded reasoning.

Class	A: Claim only	B: Gold evidence	C: BM25 top-5
CONTRADICT	0.0476	0.0444	0.0659
NOT ENOUGH INFO	0.5865	0.8442	0.5616
SUPPORT	0.4929	0.6078	0.4676

Table 2. Per-class F1 on all 286 matched development claims.

7.3. Empty-evidence shortcut diagnostic

A deterministic rule that predicts NOT ENOUGH INFO whenever condition-B evidence is empty and SUPPORT otherwise achieves accuracy 0.7902 and macro-F1 0.6009 on all 286 claims. Both metrics exceed condition B on the full set (accuracy 0.6014; macro-F1 0.4988). This simple rule uses evidence presence as a proxy for the label, revealing that the all-claim B comparison is structurally confounded.

On the 182-claim gold-evidence slice, the same rule predicts SUPPORT for every claim and achieves accuracy 0.6703 and macro-F1 0.4013, again exceeding condition B (accuracy 0.4835; macro-F1 0.3492). The B-minus-rule macro-F1 difference on this slice is -0.0521 (95% CI: -0.0927 to -0.0057). These results caution against interpreting condition B performance as evidence that the classifier learned to interpret scientific evidence.

7.4. RQ2: retrieval and downstream correctness

BM25 retrieves at least one annotated gold evidence document for 164 of the 182 eligible claims, giving a claim-level Hit@5 of 90.11%. TF-IDF cosine retrieval does so for 151 of 182 claims, giving 82.97%. These metrics describe whether a claim has at least one gold-document hit among the top five results; they do not measure the fraction of all gold documents retrieved.

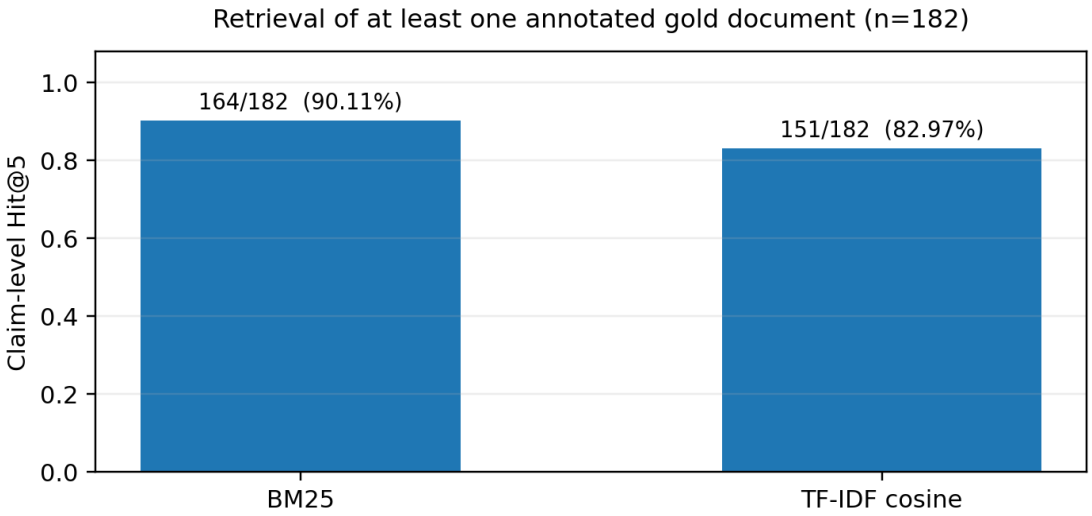


Figure 2. Claim-level Hit@5 over the 182 claims with annotated gold evidence. A hit requires at least one annotated gold document among the top five retrieved documents.

Despite BM25's higher Hit@5, condition C does not outperform claim-only condition A on the 182-claim evidence slice: C accuracy is 0.3736 compared with 0.3901 for A. The hit/miss breakdown is shown in Table 3. Among the 164 BM25 hit claims, the class composition is 115 SUPPORT and 49 CONTRADICT; among the 18 misses, it is 7 SUPPORT and 11 CONTRADICT. The corresponding accuracies differ across groups, but the small miss group and different class composition make the comparison descriptive rather than causal.

BM25 status	n	Class composition	A accuracy	B accuracy	C accuracy
Hit	164	115 SUPPORT; 49 CONTRADICT	0.4024	0.5000	0.3963
Miss	18	7 SUPPORT; 11 CONTRADICT	0.2778	0.3333	0.1667

Table 3. Descriptive accuracy by BM25 hit/miss status on the 182-claim evidence slice.

The recorded Fisher exact test for condition C hit versus miss has p = 0.0723. Given the 18-claim miss group and the descriptive nature of this subgroup comparison, the result is not treated as a definitive hypothesis test or as evidence of a causal relationship.

7.5. Sensitivity to the near-duplicate threshold

At a token-set Jaccard threshold of 0.70, the sensitivity analysis contains 209 matched claims and 127 evidence-bearing claims. Within the evidence-bearing subset, B-A is +0.0564 (95% CI: -0.0401 to 0.1535), and C-A is -0.0037 (95% CI: -0.0901 to 0.0852). Both intervals cross zero. These estimates are directionally compatible with the primary evidence-slice results but remain uncertain. Because the threshold changes the included claims and class distribution, this is a sensitivity analysis rather than a replication on the same sample.

Sensitivity analysis (Jaccard 0.70)	Estimate	95% bootstrap CI
B-A macro-F1 difference	+0.0564	-0.0401 to 0.1535
C-A macro-F1 difference	-0.0037	-0.0901 to 0.0852

Table 4. Paired macro-F1 differences on the 127-claim sensitivity evidence slice.

8. Discussion

The results support three main observations. First, condition B appears substantially stronger on the full three-class matched set, but that comparison is confounded by the empty-evidence representation of NOT ENOUGH INFO claims. A deterministic evidence-presence rule outperforms the B classifier on the full set, making it inappropriate to interpret the aggregate gain as improved evidence interpretation. Second, on claims with annotated gold evidence, the B-A point estimate is positive but its paired confidence interval includes zero. The predeclared criterion for a reliable practical improvement is therefore not met. Third, BM25 retrieves at least one annotated gold document for more eligible claims than TF-IDF cosine retrieval, but this advantage does not translate into higher classification accuracy for condition C relative to claim-only condition A.

The extremely low CONTRADICT F1 across all three conditions is consistent with a limitation of bag-of-words representations. One possible explanation is that lexical features do not adequately encode negation, directionality, or relational context. However, the present experiment does not isolate this mechanism, so it should be treated as a plausible interpretation rather than an experimentally established cause.

A second possible issue is the amount of text supplied to condition C. The project audit reports approximately 8,000 characters on average for the concatenated retrieved input and indicates that the 100,000-feature cap binds. Input length and feature saturation may affect performance, but their individual effects were not isolated in the archived experiments. A controlled ablation would be required before making a causal claim about either factor.

Taken together, the findings justify a cautious negative or inconclusive conclusion rather than a performance claim. Retrieval is useful for identifying candidate documents, as measured by claim-level Hit@5, but the lexical classifier may still fail to use those documents to produce the correct verdict. Retrieval quality and label correctness should therefore be measured and reported independently.

9. RQ3 Status: Preliminary Error Review

A stratified sample of 50 incorrect claim-condition predictions was drawn after a seven-category codebook had been frozen and a post-draw amendment had been recorded for claims without gold evidence. The sample contains 17 errors from A, 17 from B, and 16 from C, selected using seed 42. A separate AI-assisted preliminary review assigned one provisional code to each row after inspecting claim text and relevant gold or retrieved abstract text obtained from public SciFact mirrors.

The preliminary file contains 31 LABEL DECISION ERROR assignments, 14 AMBIGUOUS ANNOTATION assignments, three OTHER UNCERTAIN assignments, one NO EVIDENCE SEEN assignment, and one RETRIEVAL MISS assignment. The two metadata-deterministic assignments are B/303 (NO EVIDENCE SEEN) and C/756 (RETRIEVAL MISS); the remaining 48 are text-review judgments. These counts describe the provisional coding file only.

The public text mirrors used for the preliminary review are not checksum-matched to the exact historical local corpus snapshot used in the experiment. All 50 labels therefore require student/domain verification against that local snapshot and the condition-specific inputs. The counts are not final human-coded prevalence estimates, are not estimates for the full SciFact dataset, and are not presented as final taxonomy findings. No independent second annotator was used, so no inter-annotator agreement statistic or Cohen’s kappa is reported. Raw abstracts are not redistributed; the project records source URLs, identifiers, uncertainty notes, and a student verification sheet. The original frozen sample and codebook remain unchanged, and the preliminary labels are retained as a separate versioned artifact.

10. Limitations and Threats to Validity

The study uses one development split and a single frozen run rather than an untouched labelled test set or a multi-seed estimate. The matched set contains 286 claims after near-duplicate filtering and is smaller than the full development split. The three conditions are separately fitted models from a common model family; the design does not isolate a causal intervention on one fixed model. In addition, the empty-evidence shortcut in condition B confounds the full-set comparison, and the evidence-bearing slice has only two classes.

The lexical verifier has very low CONTRADICT F1 in every condition. The BM25 miss group contains only 18 claims, limiting subgroup interpretation. BM25 epsilon behaviour for frequent terms and the effect of the feature cap are not isolated. The condition-B evidence construction selects the first rationale set per document, and 75 evidence-eligible claims have multiple rationale sets in at least one document. Results could differ under a union-of-rationales or sentence-level aggregation protocol.

RQ3 coding remains preliminary. All 50 sampled rows have AI-assisted provisional codes, but the text sources are public mirrors that are not checksum-matched to the historical corpus; the student must verify every row against the exact local snapshot and the corresponding model inputs. No independent second coder was used. Therefore, neither final taxonomy claims nor inter-annotator agreement can be reported.

The archived project record does not establish a complete clean-checkout reproduction. Colab/GPU execution was not verified. The original environment snapshot records Python 3.11.2 and CPU-only Torch, with CUDA unavailable. The independent review container used Python 3.13.5 and did not have rank-bm25 installed, preventing collection of two retrieval-dependent test modules. These environment and test limitations are detailed in Section 12.

Finally, the official SciFact repository license file identifies CC BY 4.0 for claim/evidence annotations, ODC-By 1.0 for corpus abstracts, and Apache 2.0 for code [12]. However, the downloaded source was a moving release/latest archive rather than an immutable commit, while a Hugging Face mirror displays a different CC BY-NC 2.0 license label and a different split layout. The exact association between the downloaded archive and an upstream revision is therefore not fully resolved. Raw data remain excluded pending snapshot-level license confirmation.

11. Conclusion

This matched SciFact development-set study compared claim-only, annotated gold-evidence, and BM25-retrieved-evidence inputs using separately fitted TF-IDF and logistic-regression classifiers. The gold-evidence condition achieved the highest macro-F1 on the full 286-claim set, but this result is confounded by empty-evidence NOT ENOUGH INFO examples and is exceeded by a simple evidence-presence rule. On the 182-claim gold-evidence slice, the estimated B-A macro-F1 gain is +0.0352, but its 95% paired-bootstrap interval crosses zero and does not satisfy the predeclared improvement rule.

BM25 achieves a claim-level Hit@5 of 90.11%, compared with 82.97% for TF-IDF cosine retrieval, yet the BM25-based classifier does not improve over the claim-only baseline on the evidence-bearing slice. The most defensible conclusion is therefore that retrieval success alone does not ensure correct scientific-claim classification under the tested lexical pipeline. The archived comparison also exposes a structural shortcut that must be addressed in future evidence-conditioned evaluation. A final taxonomy-guided error analysis remains future work until all 50 provisional codes are verified against the exact historical corpus snapshot; an independent second coder would be required to report agreement.

12. Reproducibility and Artifact Traceability

The companion project repository contains source modules, scripts, YAML configuration, prediction CSV files, retrieval-rank files, metric JSON, confusion matrices, the error codebook, sampling metadata, protocol-change log, and original run logs. Prediction files are timestamped, and their SHA-256 hashes are embedded in the evaluation metadata. A read-only verifier in the final package checks those hashes, the 286-row count, unique claim identifiers, and equality of the A/B/C claim sets. The final review reran this verifier successfully. It did not rerun model training because raw SciFact files are not included in the project archive.

The original environment snapshot records Python 3.11.2 and CPU-only Torch; CUDA was unavailable. The project does not provide evidence of Colab execution. The independent review container used Python 3.13.5 and did not have rank-bm25 installed. The full test suite could not complete collection because two retrieval-dependent test modules require rank-bm25==0.2.2. Excluding only those two modules, 51 tests passed; three RQ3 enrichment tests and five metadata-code tests also passed independently. This is partial test coverage in Python 3.13.5, not a full-suite pass in the recorded Python 3.11.2 environment.

The archived project record reports a 59-test pass, but complete raw test stdout was not included in the source archive. That historical count is therefore project-reported and was not independently reproduced in this review. The upstream acquisition setting points to a moving release/latest tarball. File hashes are recorded, but the upstream commit identifier is

not. The component-level license terms were identified from the official AllenAI repository: CC BY 4.0 for claims/evidence annotations, ODC-By 1.0 for S2ORC abstracts, and Apache 2.0 for repository code [12]. Because the downloaded archive is not mapped to an immutable upstream commit, archive-to-revision license association remains partially unresolved. A fresh clean-checkout reproduction, an immutable upstream revision pin, and snapshot-level license confirmation remain necessary before any raw-data redistribution.

Data and Code Availability

The code, scripts, configurations, archived result outputs, and logs are included in the accompanying project repository. Raw SciFact files and fitted model objects are omitted. The recorded acquisition source is the official AllenAI SciFact S3 release/latest archive, downloaded on 30 October 2026; the prior project log records its SHA-256 hash. The official repository license assigns CC BY 4.0 to claims/evidence annotations, ODC-By 1.0 to S2ORC corpus abstracts, and Apache 2.0 to code [12]. The downloaded archive is identified by its recorded hash but is not mapped to an immutable upstream commit. A separate Hugging Face mirror lists CC BY-NC 2.0 and a different split layout; it was not the source used for these experiments. Raw data are excluded pending snapshot-level confirmation.

AI Assistance and Research Integrity

AI tools supported coding, debugging, and document preparation, as disclosed in the project's AI assistance log. Numeric results in this manuscript are taken from timestamped project outputs; no new model-training or retrieval experiment is claimed in this revision. An AI-assisted preliminary text review of all 50 RQ3 sample rows is documented separately. Those codes are not student-final and require verification against the exact historical local corpus. The conceptual link between the earlier literature review and this experiment is documented in a separate project traceability note; the student remains responsible for reviewing and owning that interpretation.

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